

Quiz 5

1 FILL IN THE BLANK

0.5 / 0.5



Ackermann steering is a type of wheeled locomotion mechanism similar to car steering where the front wheels turn to steer.

2 TRUE/FALSE

0.5 / 0.5



Tracked locomotion provides high speed on rough terrain.

☐ True

☒ False

3 MULTIPLE CHOICE

0.5 / 0.5



Which of the following is an advantage of legged locomotion?

☐ A. High speed on flat surfaces

☒ B. Ability to navigate rough and uneven terrain

☐ C. Simple control mechanisms

☐ D. Low energy consumption

4 TRUE/FALSE

0.5 / 0.5



Continuous tracks in tracked locomotion provide continuous contact with the ground, allowing the robot to traverse over obstacles.

☒ True

☐ False

5 FILL IN THE BLANK

0.5 / 0.5



Undulatory motion in aquatic locomotion mimics the movement of fish.

6 MULTIPLE CHOICE

0 / 0.5



What is locomotion in robotics?

- ☐ A. The process of powering a robot.
- ☐ B. The sensory perception of a robot.
- ☐ C. The mechanisms that enable robots to move from one place to another.
- ☒ D. The programming of robot behaviors.

7 FILL IN THE BLANK

0.5 / 0.5



The process of guiding a robot along a planned path while avoiding obstacles is known as navigation.

8 TRUE/FALSE

0.5 / 0.5



Model Predictive Control (MPC) is a control strategy that optimizes control inputs by predicting future states of the system using a model.

☒ True

☐ False

9 FILL IN THE BLANK

0.5 / 0.5



The PID control strategy uses feedback to maintain the desired output by adjusting control inputs based on proportional, integral, and derivative gains.

10 MULTIPLE CHOICE

0.5 / 0.5



Which type of aerial locomotion mechanism uses wings for lift and propellers for thrust?

☐ A. Drones

☒ B. Fixed-Wing Aircraft

☐ C. Hovercraft

☐ D. Multirotors